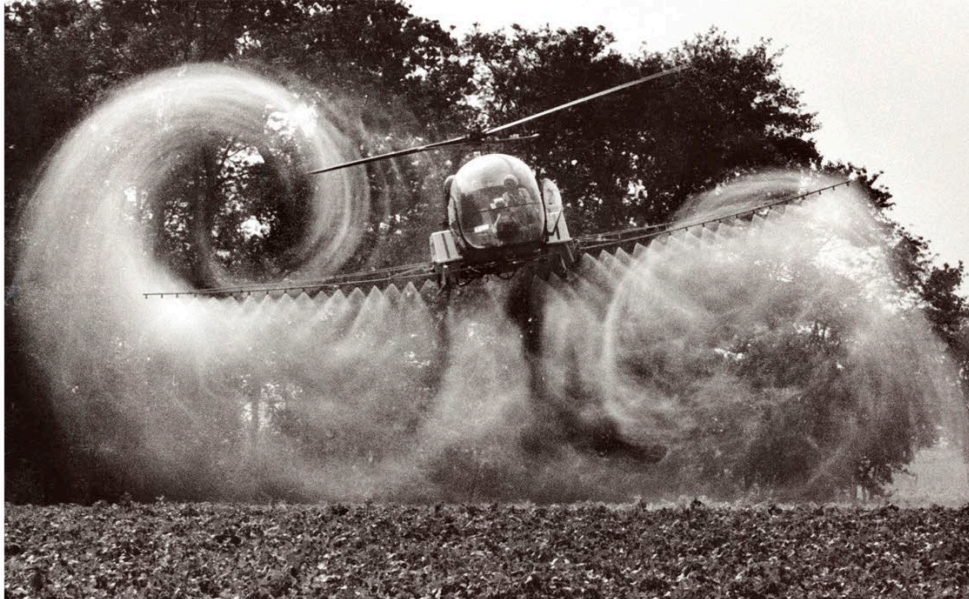




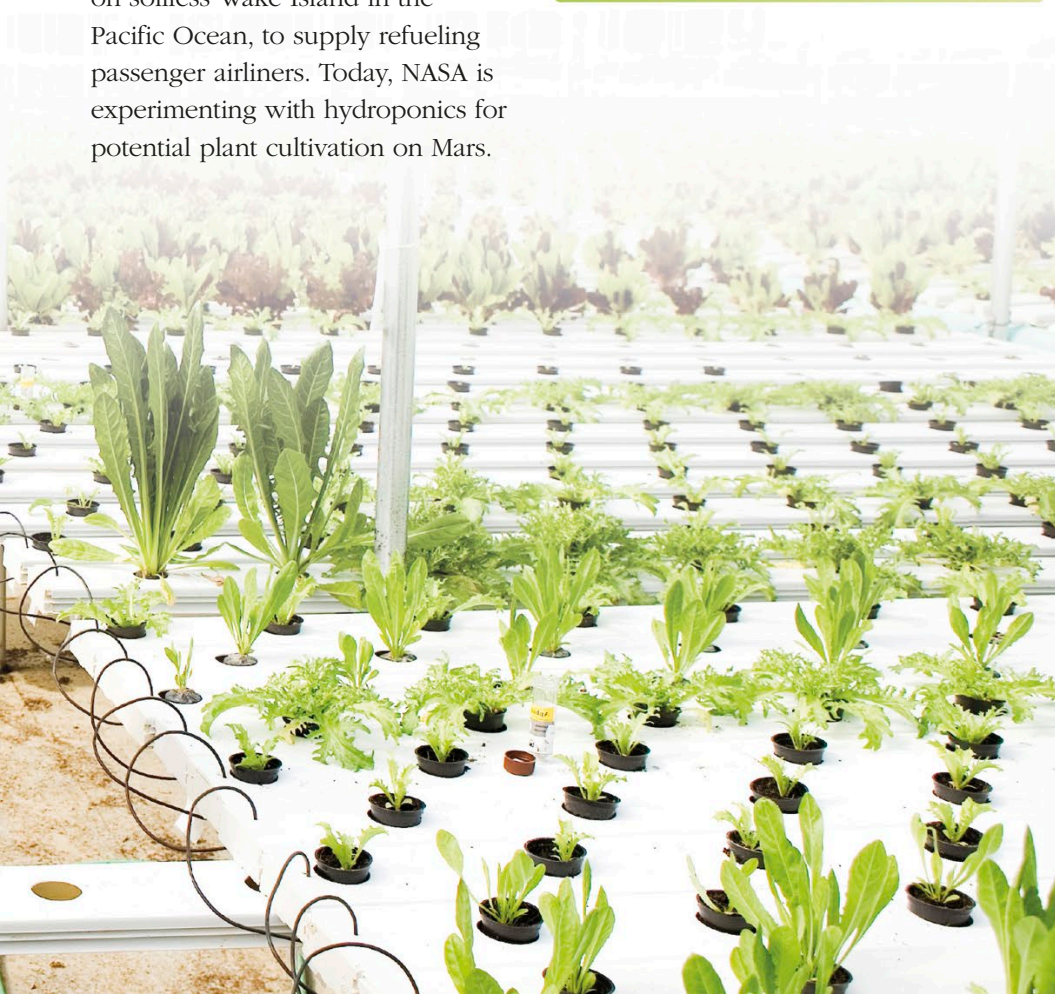
A helicopter sprays potatoes with pesticide in the UK

HYDROPONICS

In 1929, American researcher William Gericke grew tomato vines 25 ft (7.6m) long using just high-nutrition minerals mixed with water. This way of growing plants without soil became known as hydroponics. In the 1930s, vegetables were grown this way on soilless Wake Island in the Pacific Ocean, to supply refueling passenger airliners. Today, NASA is experimenting with hydroponics for potential plant cultivation on Mars.

WOW!

Nearly 30 percent of the global population works in agriculture, making it the single largest industry in the world.



PEST CONTROL

In 1939, Swiss chemist Paul Müller discovered that a chlorine-based chemical, commonly called DDT, killed insects but had little effect on warm-blooded animals. DDT was used widely in agriculture for years, but today it has been replaced by more effective and safer pesticides.



FERTILIZERS

In 1909, German chemist Fritz Haber succeeded in capturing nitrogen from the air to form ammonia, which could be made into plant fertilizers. Another German chemist, Carl Bosch, developed this method for large-scale ammonia production in massive factories (above). Since then, this breakthrough has led to a huge increase in food production globally.



GM CROPS

In 1969, American biochemists discovered how to alter living things (to add more flavor, for example) by splicing the genes of one organism into another. The first genetically modified (GM) food approved for sale was the Flavr Savr tomato (above), produced by the US company Calgene in 1994.